

Basic Principles of Medicine 1

Module: FTM

Membrane structure and transport:
Integrated Basic principles
DLA

Dr. Mohamed Idries
midries@sgu.edu

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Overview: The FTM 1 module discusses the following aspects of each discipline as it relates to the Fundamental concepts of Molecular Medicine

DISCIPLINE	CONCEPTS DISCUSSED
HISTOLOGY	• Function of proteins associated with cellular membranes • Structure of the proteins of cellular membranes • Technique to visualize the protein structure of cellular membranes • Demonstration of the link between membrane proteins and their role as receptors • Understand membrane permeability for lipophilic and hydrophilic compounds
BIOCHEMISTRY	• Cell structure and components of cell membrane • Molecular Biology and central dogma • Proteins, Enzymes • Signal transduction

Objectives: DLA: Membrane Structure and Transport: Integrated basic principles

SOM.MK.I.BPM1.1.FTM.3.BCHM.0010	Describe the structure, composition and significance of the fluid mosaic model of cell membrane
SOM.MK.I.BPM1.1.FTM.3.BCHM.0011	Review the lipid (cholesterol, phospholipids and glycolipids) composition of the plasma membrane. Discuss the relevance of cholesterol and lipid composition in regulating membrane fluidity
SOM.MK.1.BPM1.1.FTM.3.HCB.0683	Differentiate between the two (2) major classes of membrane proteins.
SOM.MK.1.BPM1.1.FTM.3.HCB.0684	List the two major categories of integral membrane proteins.
SOM.MK.1.BPM1.1.FTM.3.HCB.0685	Describe the six major functions of the two (2) major categories of integral membrane proteins.
SOM.MK.1.BPM1.1.FTM.3.HCB.0686	Discuss membrane permeability in relation to membrane composition, and hydrophilic and lipophilic compounds
SOM.MK.I.BPM1.1.FTM.3.BCHM.0014	Differentiate between active and passive transport systems based on energy requirement and movement along the concentration gradient.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0015	Discuss facilitated diffusion by glucose transporters (GLUT 1-5) highlighting their location and characteristics in relation to substrate affinity and insulin sensitivity.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0017	Distinguish between primary active transport (Na^+/K^+ -ATPase) and secondary active transport (SGLT).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0018	Describe the CFTR channel as an example of ABC transporters. Predict effects of CFTR mutations on lung and pancreas.

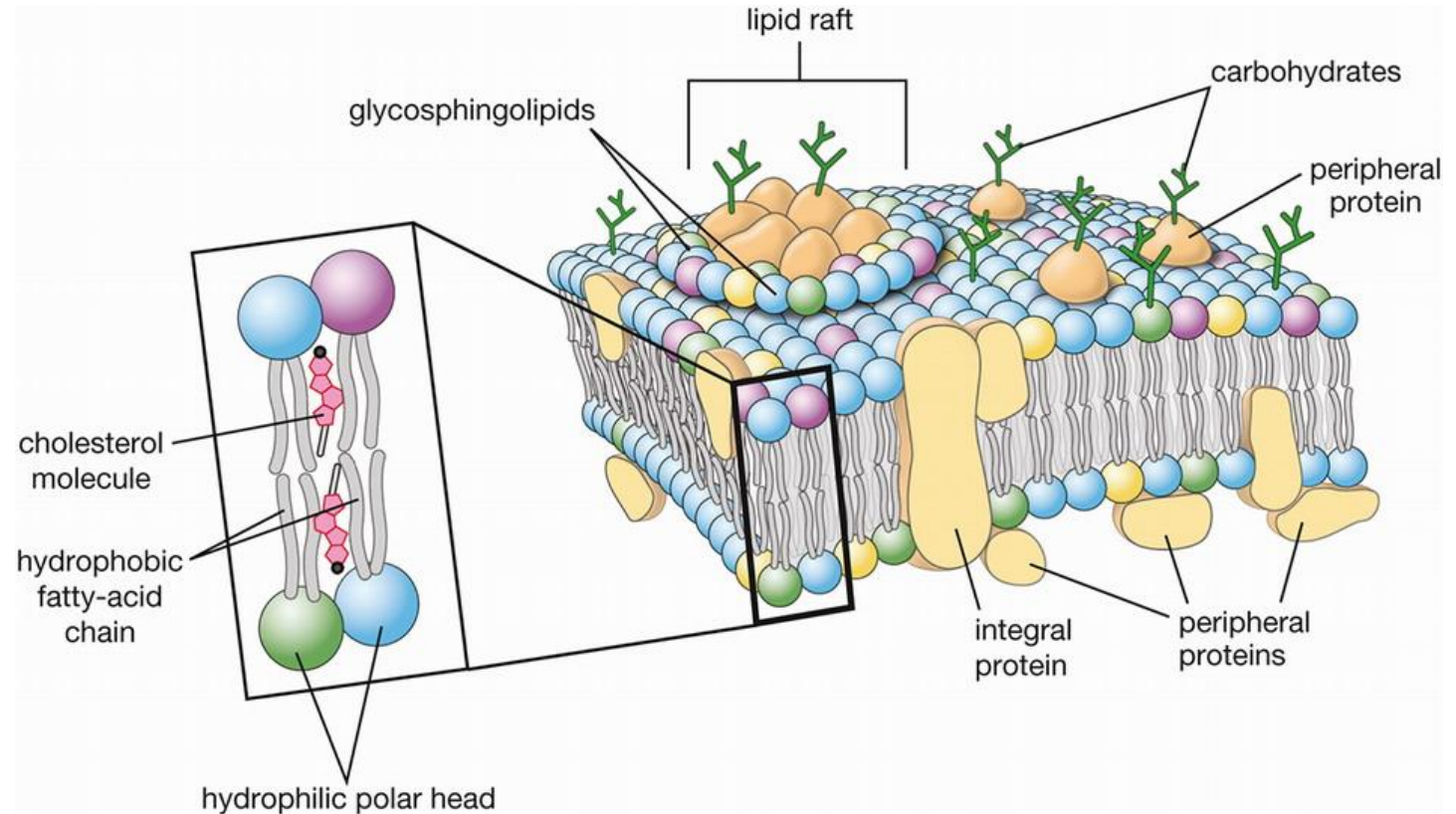
Recommended Reading

- **Mark's Basic Medical Biochemistry: A clinical approach (MBMB)** 6th edition Chapter 10: Section II. Plasma membrane; [Cell Structure and Signaling by Chemical Messengers | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library \(lwwhealthlibrary.com\)](#)
- **Lippincott Illustrated Reviews (Lippincott) Biochemistry** 9th edition; Chapter 7; Section: Intestinal absorption of monosaccharides: Pages 96 (Fig 7.10), Chapter 8: Section: Glucose transport into cells;
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- **LIPPINCOTT'S ILLUSTRATED REVIEWS: CELL AND MOLECULAR BIOLOGY**, N. CHANDAR & S. VISELLI, 2ND ED, Units I & III; <https://meded-lwwhealthlibrary-com.periodicals.sgu.edu/book.aspx?bookid=2335>

Fluid Mosaic Model

Mammalian cell membrane:

- Phospholipid bilayer
- Embedded proteins
- Dynamic, fluid-like structure
- Selective barrier



The membrane is referred to as a **fluid mosaic** because it contains proteins and lipids that mostly move laterally within its plane.

Modified Fluid Mosaic Model

- Plasma membrane composed of **lipid rafts**.
- **Lipid rafts:**
 - Specialized, patchy, localized microdomains; unique protein and lipid composition.
 - Distinct in structure and function.
 - High concentration of cholesterol and longer, highly saturated fatty acid chains.

- Two types of rafts:

Planar Rafts

- Short lived.
- Planar.
- Contain **Flotillins**

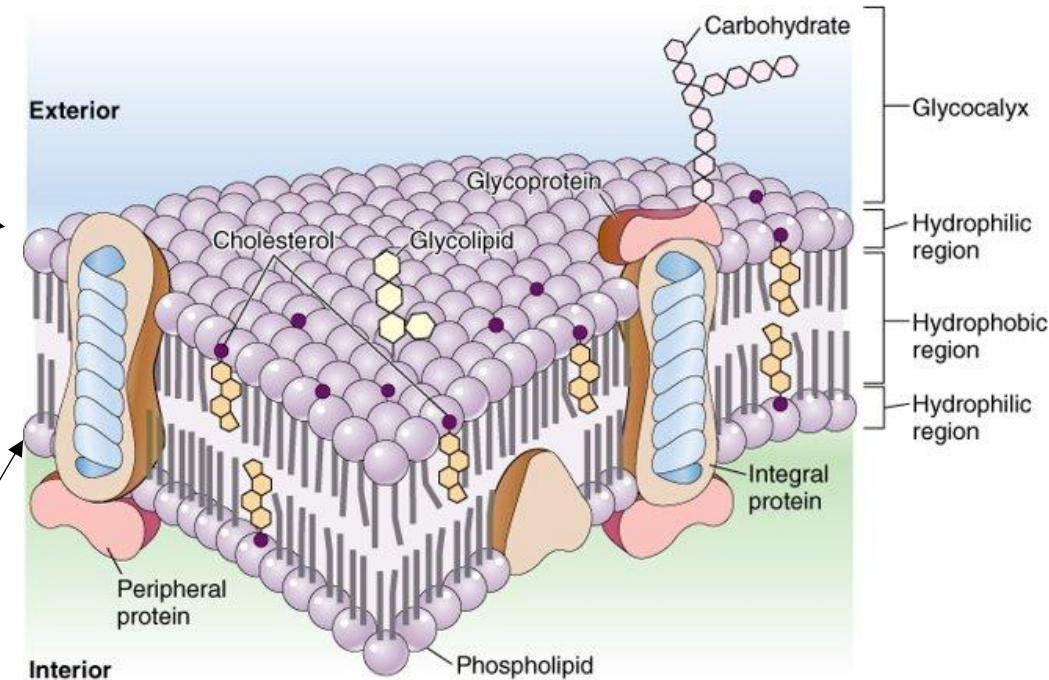
Caveolar Rafts

- Long lived.
- Invaginations.
- Contain **Caveolins (signal transduction)**.
- Prominent in **smooth muscle cells**.

Structure of Plasma Membrane

Outer membrane (leaflet):

- **Phosphatidylcholine** (mainly).
- **Sphingomyelin** (mainly).
- Phosphatidylethanolamine (less common).
- **Glycoproteins** and **glycolipids** (proteins and lipids covalently bound carbohydrate chains) to form the **glycocalyx**.



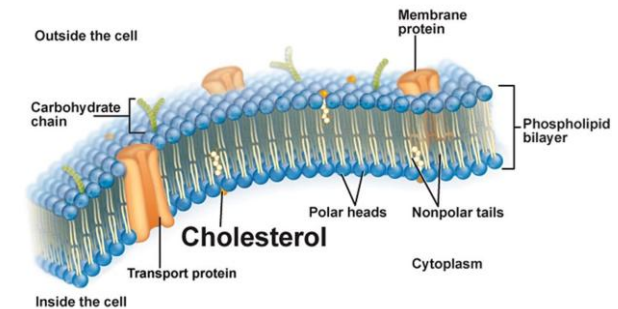
Inner membrane (leaflet):

- **Phosphatidylethanolamine** (negatively charged).
- **Phosphatidylserine** (when it flips to the **outer leaflet** → **signal for apoptosis**).
- **Phosphatidylinositol** (**only in the inner membrane**, can be used as **PIP₂** in **calcium messenger system**).

Structure of Plasma Membrane

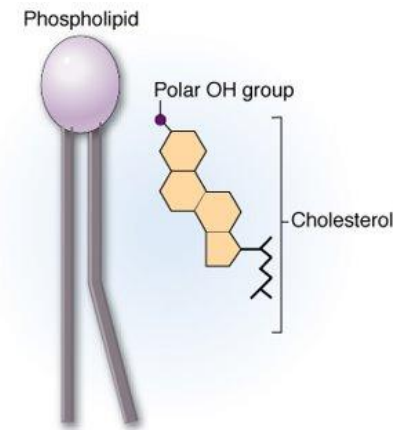
What is the role of cholesterol in plasma membrane?

- Free cholesterol is found in the **plasma membrane**, but **not in** intracellular membranes.
- **Maintains membrane fluidity.** A specific membrane fluidity is needed for membrane proteins to function.



How is the fluidity of a plasma membrane determined?

- The **fatty acid composition** of the membrane lipids
 - Saturated fatty acids decrease fluidity
 - Higher the number of double bonds, greater the fluidity
- The amount of **free cholesterol**
 - Cholesterol prevent hydrophobic chains from packing too closely together.



Structure of Plasma Membrane

Cholesterol prevents drastic changes in plasma membrane fluidity during temperature changes

Low Temperature



The **fatty acyl** groups are **stiff** and cholesterol prevents decrease of membrane fluidity by separating acyl chains and preventing too close packing.

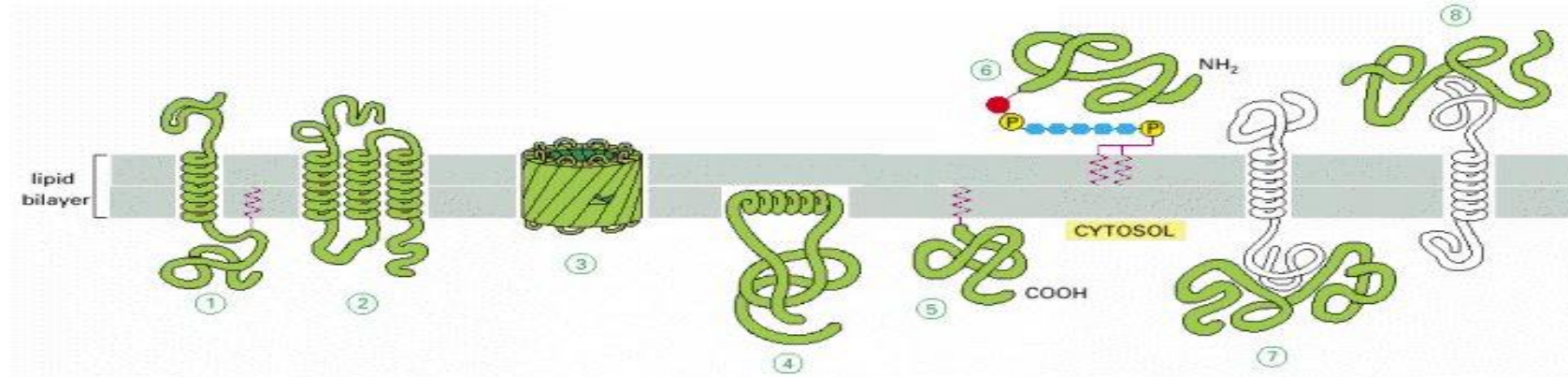
High Temperature



The **fatty acyl** groups are **fluid** and cholesterol prevents increase of membrane fluidity and slows down the movement of the esterified fatty acids.

Structure of Plasma Membrane

Membrane proteins



Integral Membrane Proteins (6 General Categories)

- Transmembrane (1 – 3)
- Anchored (4 – 6)
 - Alpha helix
 - Lipid chain
 - Oligosaccharide linker

Peripheral Membrane Proteins

- Noncovalent association with integral membrane proteins (7 – 8)

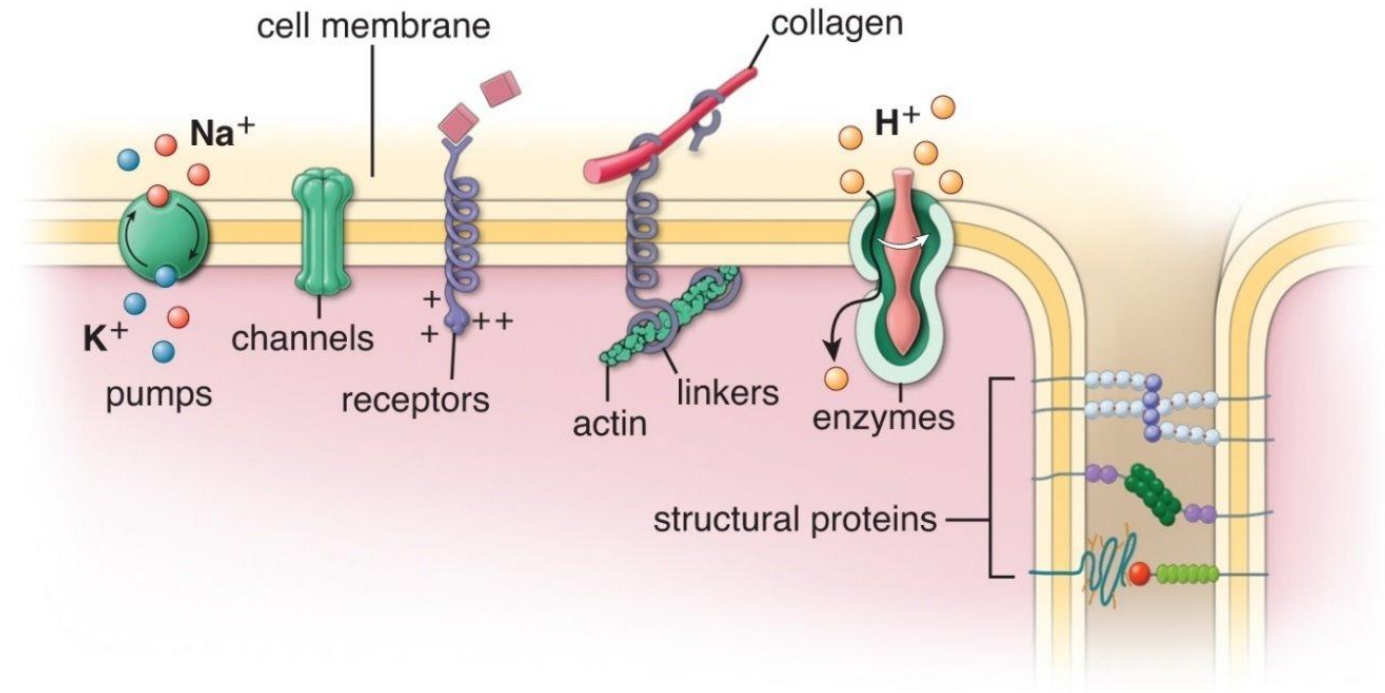
Functions of Plasma Membrane Proteins

1. Pumps/Carriers/Transporters:

- Transport specific ions across the membrane (e.g., Na^+ , K^+).
- Transport metabolic precursors (e.g., amino acids, sugars).

2. Channels:

- Transport of ions, small molecules and water (passive diffusion).
- Aquaporins (water).
- Gap junctions (passage of ions, small molecules between adjacent cells).



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3. Receptors

4. Linkers

5. Enzymes

6. Structural Proteins

Plasma Membrane Permeability

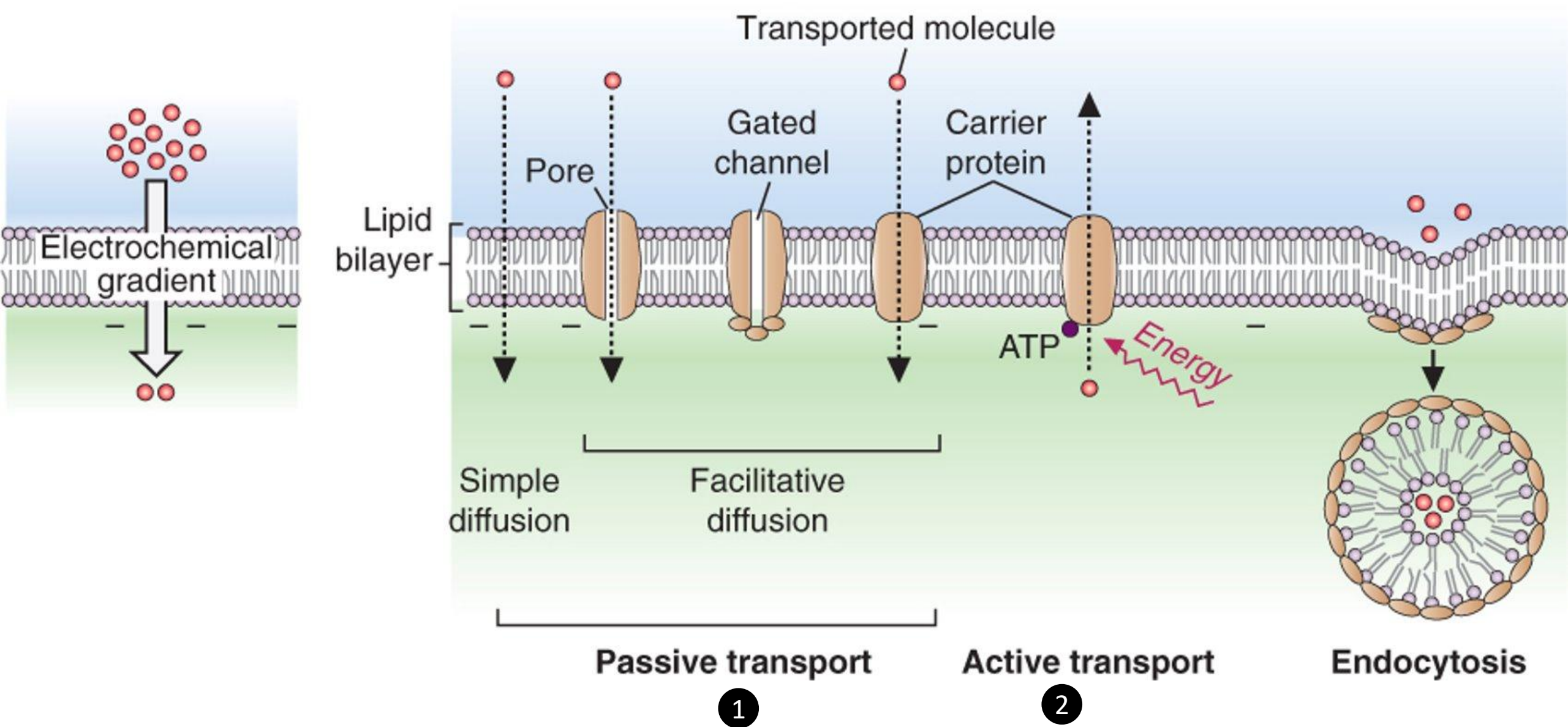
Plasma membrane has **selective permeability**

Permeable	Less Permeable	Highly Impermeable
Hydrophobic, non-polar, uncharged	Small, polar, uncharged	Charged (ions), polar
Examples: O ₂ , CO ₂	Example: H ₂ O	

Channels and Transporters are **transmembrane proteins** that increase the permeability of the membrane to molecules that can't cross.

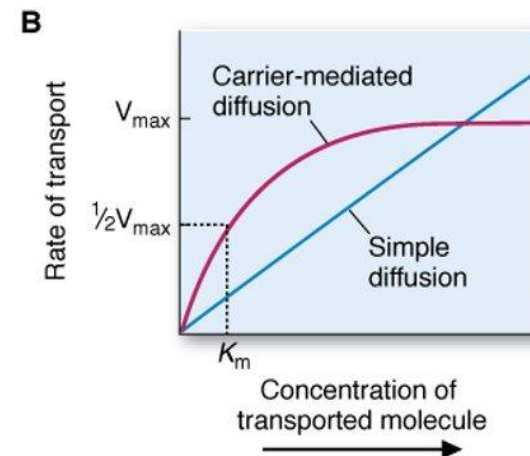
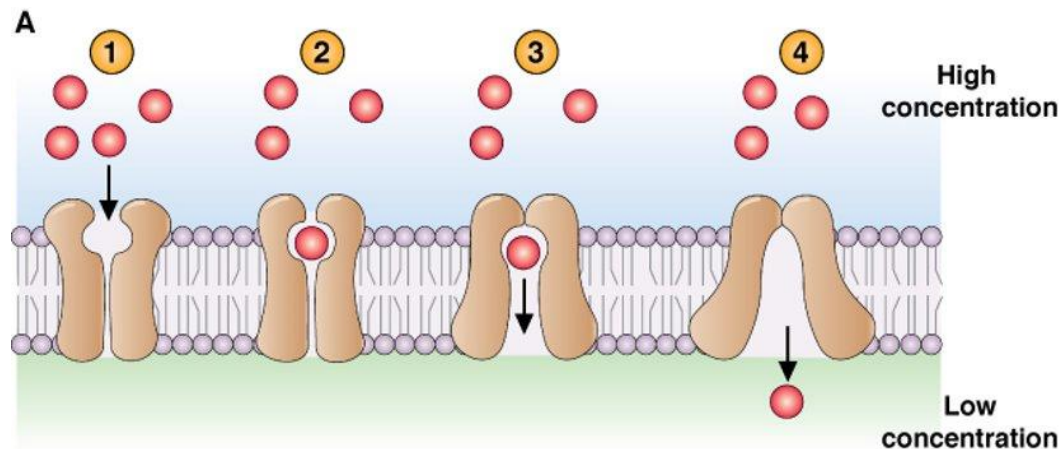
Common types of transport mechanisms include **passive** and **active transport**:

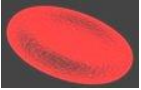
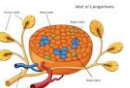
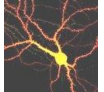
- Active transport: requires energy (e.g., Na⁺/K⁺ pump)
- Passive transport: do not require energy



Passive Transport

- No energy is required.
- **Facilitated diffusion:**
 - Transporter facilitates diffusion down a molecule's concentration gradient.
 - Molecule transported from area of high concentration to an area of low concentration.
 - Process is **specific** and **saturable**.
 - It has a **maximal transport curve**
 - Example: **GLUT (glucose transporter)** family of transporters (different isoforms)



	Location/Tissue distribution	Note
GLUT-1	- Most tissues. - Abundant in RBC and blood-brain barrier.  	- High affinity glucose transporter - Basal glucose uptake is at a constant rate at both normal and lower blood glucose concentrations.
GLUT-2	- Liver - Pancreatic beta-cells - Kidney - Basolateral side of intestinal mucosa cells   	- High-capacity, low affinity transporter - A blood glucose sensor for insulin release in pancreatic beta cells.
GLUT-3	- Most tissues. - Abundant in brain (neurons)  	- High affinity glucose transporter - Major transporter in the central nervous system
GLUT-4	- Fat cells. - Skeletal muscle. - Heart muscle.  	- High affinity glucose transport system that is insulin-dependent. - Insulin leads to increase of GLUT-4 on the cell surface.
GLUT-5	- Mucosal side of intestinal mucosa cells (dietary fructose). - Spermatocytes. - Kidneys.	- Preference for fructose transport and glucose transport only at very high concentration.

GLUT-2 Transporter

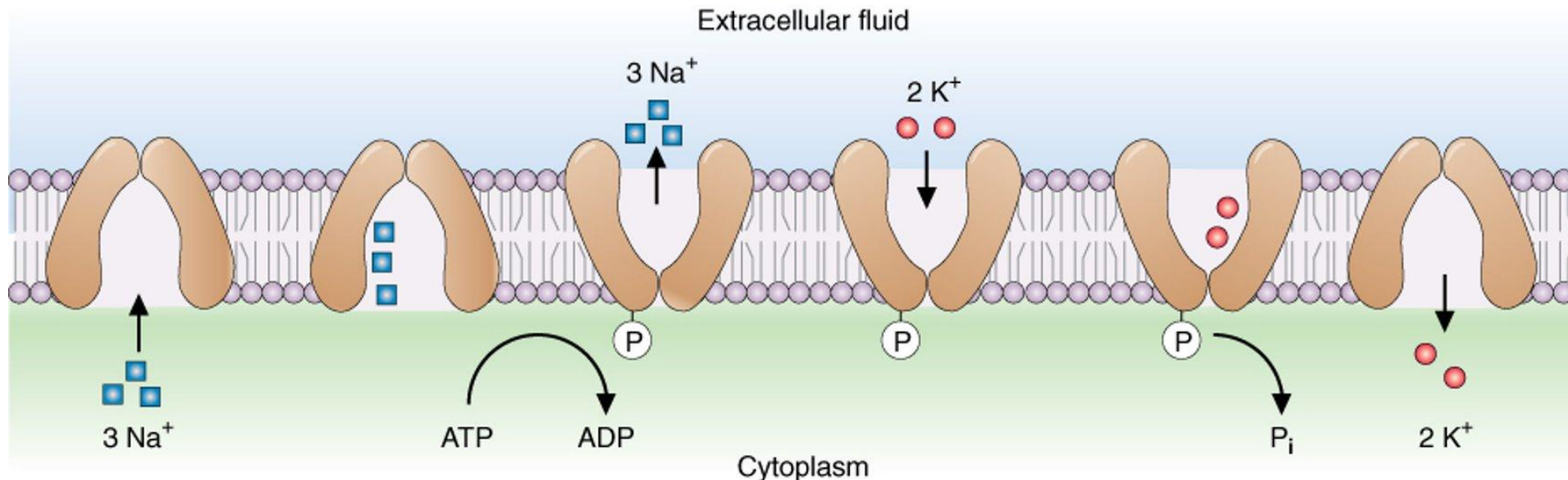
- **High capacity, low affinity** (i.e., glucose only diffuses at high concentrations).
- **Liver:**
 - **Bidirectional** transporter.
 - Transports glucose **into hepatocytes** when blood glucose is high (**after meals**).
 - Releases glucose from hepatocytes **into the blood** during low blood glucose (**fasting**).
- **Kidney:**
 - Facilitates glucose **release into the blood** during fasting.
- **Pancreatic β -cells:**
 - **Senses** glucose for **insulin secretion**.
- **Small intestine:**
 - Releases dietary monosaccharides from intestinal mucosal cells into portal system.

GLUT-4 Transporter

- Found in **adipose tissue, skeletal muscle** and heart muscle.
- **Insulin stimulates** insertion of GLUT-4 (stored in vesicles) into the plasma membrane allowing glucose uptake into these cells (**insulin dependent**).

Active Transport

- **Energy** is required.
- Energy is utilized for transport against concentration gradient.
- Example: Na^+/K^+ -ATPase pumps sodium ions out of the cell and potassium ions into the cell → high extracellular Na^+ and high intracellular K^+ .



Types of Active Transport

- **Primary active transport:**

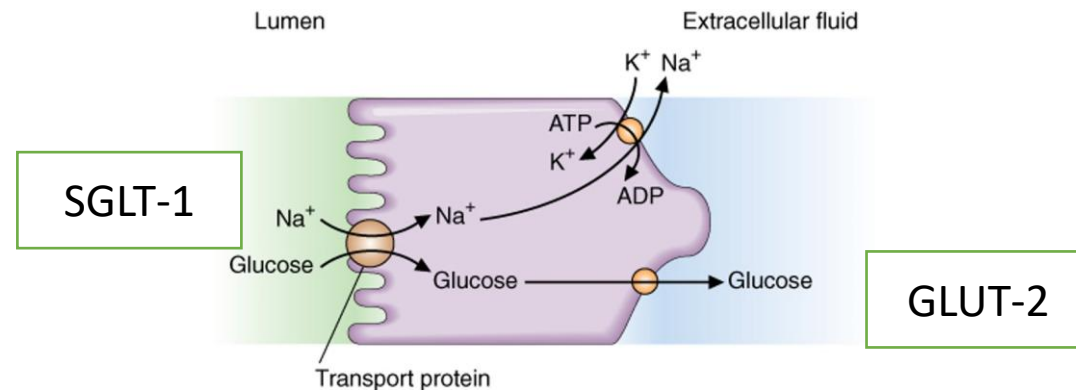
- Transport mechanisms directly coupled to the consumption of metabolic energy
- Example: **H⁺-ATPase, ATP-binding cassette transporters (ABC)**

- **Secondary active transport:**

- Refers to the translocation of a substance across a cell membrane against its concentration gradient, driven by the energy harnessed from the simultaneous movement of another substance along its concentration gradient.
- Used for the uptake of dietary amino acids, glucose or galactose into intestinal mucosal cells against their concentration gradient.
- Example: **Sodium-dependent glucose co-transporter (SGLT).**

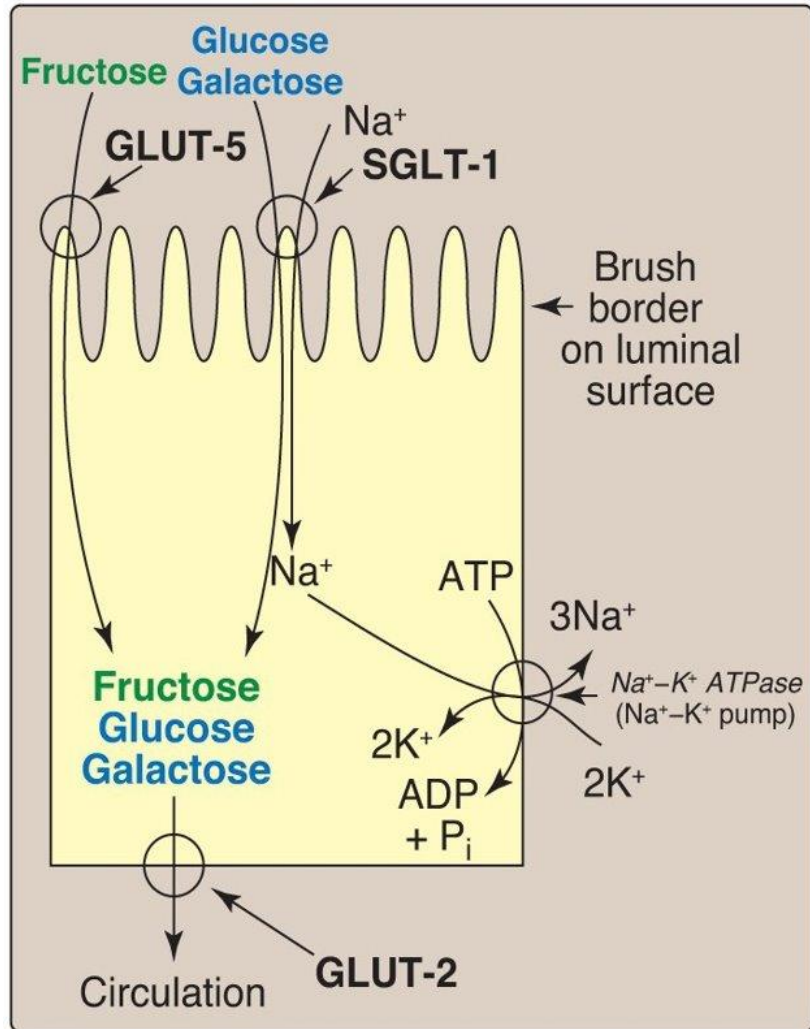
Sodium-dependent Glucose Co-Transporter (SGLT)

- Transports glucose into cells against high concentration gradient utilizing the sodium gradient maintained by the primary active transport of $\text{Na}^+/\text{K}^+-\text{ATPase}$.
- **SGLT-1**: found in **intestinal cells**
 - Dietary glucose or galactose uptake **from intestinal lumen into intestinal cells**.
 - Note: Intracellular glucose is transported from intestinal cell into blood through GLUT-2 (basolateral side).



- **SGLT-2**: found in **kidney** for glucose reabsorption from renal tubules back into blood.

Summary Dietary Uptake of Carbohydrates

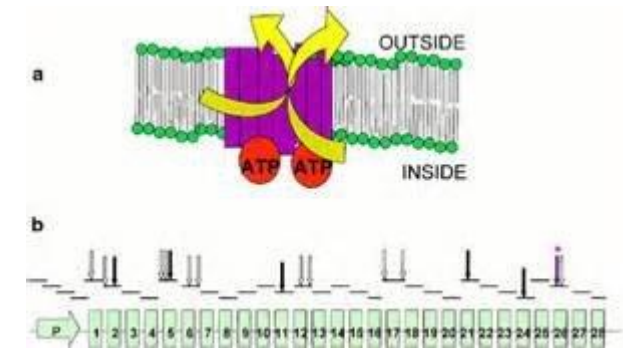
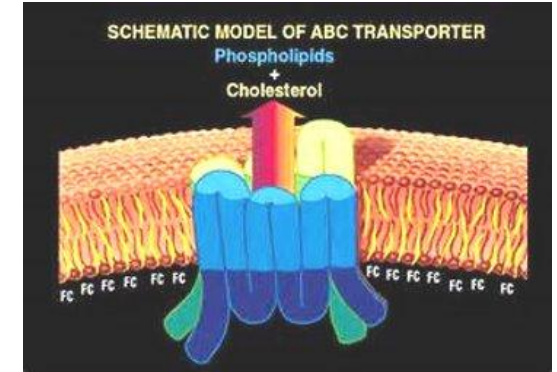


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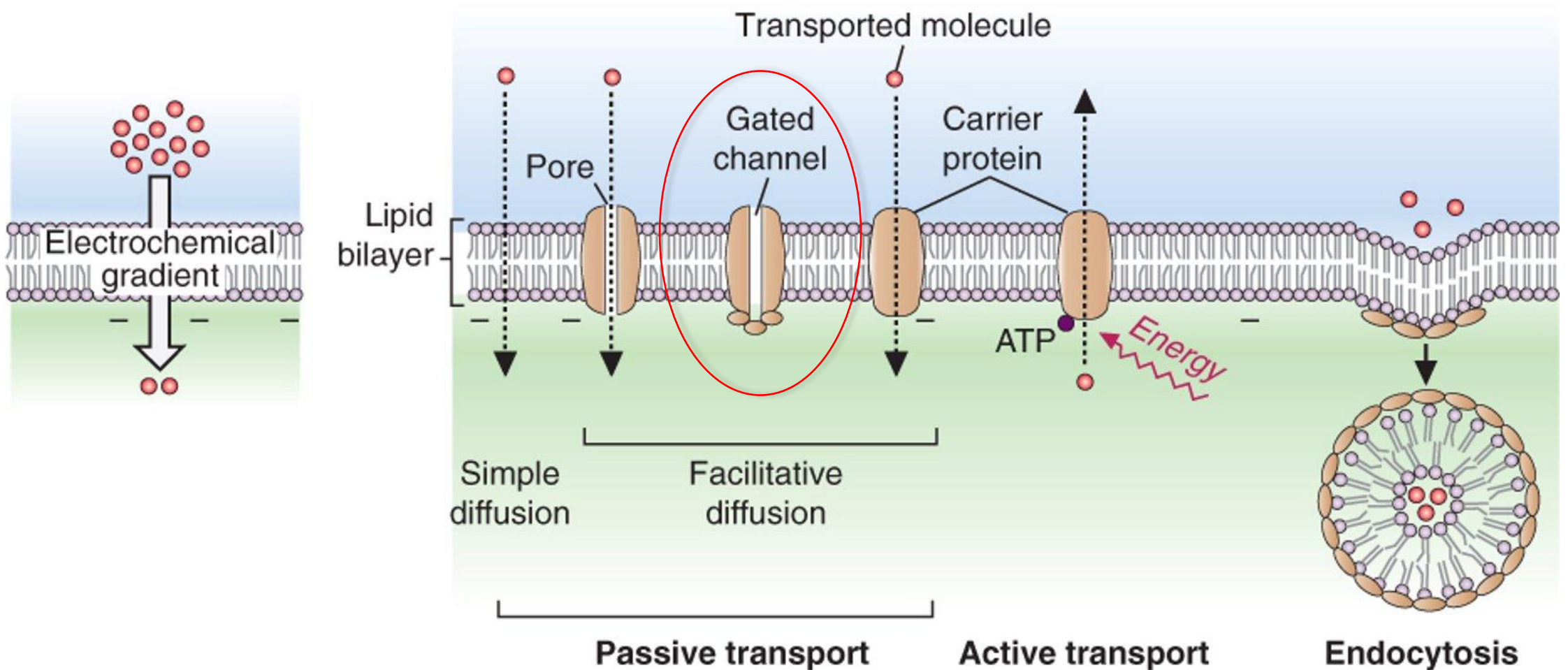
- **GLUT-5: facilitates** dietary fructose uptake.
- **SGLT-1: secondary active transport** of glucose or galactose uptake into intestinal cells from lumen. Utilizes sodium gradient (maintained by Na⁺/K⁺-ATPase).
- **GLUT-2: facilitates** release of fructose, glucose, and galactose (down their concentration gradient) from intestinal cells into blood (portal system).

ATP-Binding Cassette (ABC) Transporters

- **Primary active transport**: Binds ATP and uses the energy to drive the transport of molecules across the membrane.
- Transport lipids or lipid related compounds (e.g., phospholipids, steroids, drugs).
- Plasma membranes have ABC transporters that **transport free cholesterol** to the **outside** during plasma membrane **turnover**.
- **Liver** releases **bile salts, cholesterol, and conjugated bilirubin into bile** ducts using ABC transporters.
- A type of ABC transporter, **P-glycoprotein**, mediates **anticancer drug resistance**.
- Another type of ABC transporter, **ABCA1**, transfers cholesterol to **high-density lipoproteins**.

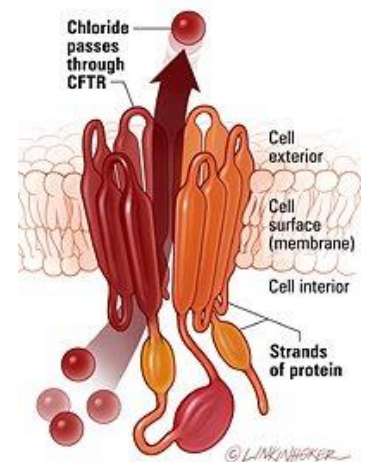
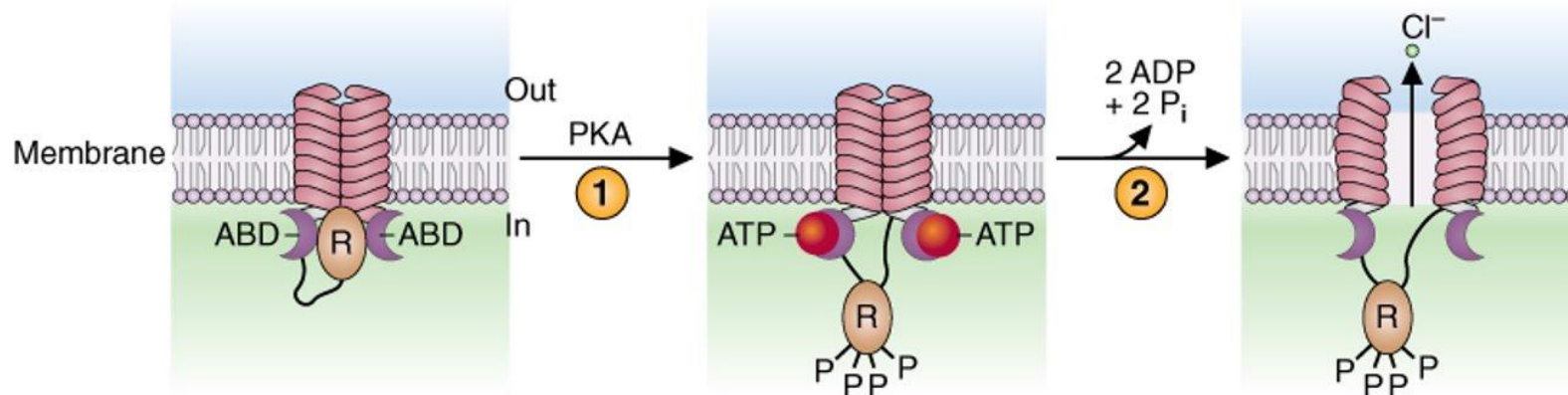


Gated Channel: Cystic Fibrosis Transmembrane conductance Regulator (CFTR)



Cystic Fibrosis Transmembrane conductance Regulator (CFTR)

- Unique type of ABC transporter.
- It is **gated chloride ion channel**.
- Uses **ATP** for **regulation** of chloride ions flux. (It doesn't use ATP for energy).
- It **does not** pump chloride; they flow with gradient once the channel opens.
- CFTR is tightly regulated, it is opened by:
 - Phosphorylation (via cAMP messenger system and protein kinase A).
 - ATP binding and hydrolysis.



Cystic Fibrosis Transmembrane conductance Regulator (CFTR)

- **Normal function:**
 - Regulates the transport of chloride and water across cell membranes
 - In the **lungs, intestines, and pancreas** (i.e., exocrine glands) CFTR transports **chloride from the cell into the airway** (lungs) and duct (pancreas, intestines) lumen → Cl^- binds Na^+ to form NaCl → attracts water into the lumen → maintains the lumen hydrated and less viscous → hydrated respiratory and intestinal mucosal linings.
 - In the **sweat glands** CFTR transports chloride **from the sweat gland ductal lumen back into the cells** → sodium follows chloride → decreases sodium lost in sweat.
- CFTR is **defective in cystic fibrosis**; either due to **absence of CFTR** or **abnormal CFTR**.

Cystic Fibrosis - Pathogenesis

- CFTR defective → decreased secretion of chloride in the intestines, pancreas, and lungs → thick sticky (hyperviscous) secretions in the intestines, pancreas, and lungs.
 - Formation of hyperviscous secretions → blockage of small passages & ducts of intestines, pancreas and lungs → chronic inflammation → tissue damage.
 - Pancreatic damage → pancreatic enzyme insufficiency → malabsorption.
 - Hyperviscous mucus in the lung airways → repeated lung infections.
- CFTR defective → inability to reabsorb chloride back from the sweat gland ductal lumen → excessive loss of water and sodium in sweat → salty sweat.

Cystic Fibrosis – Clinical Features

- Pancreatic enzyme insufficiency → malabsorption → **failure to thrive**.
- Less pancreatic lipase → malabsorption of dietary lipids → **steatorrhea** (fatty stools).
- Thick meconium (stool) obstructing terminal ileum → **meconium ileus** → newborn with bilious vomiting, abdominal distension, no passing of meconium (stool).
- Hyperviscous mucus in the lung airways → **recurrent lung infections (pneumonia)**.
- **Male infertility** due to atrophy of vas deferens.
- **Female infertility** due to thick cervical mucus.

Cystic Fibrosis – Clinical Features

- Salty sweat, why?
 - Sweat is meant to cool the skin with water and is formed with high NaCl concentrations in the coiled part of the sweat gland (blue part).
 - Chloride is reabsorbed back from the sweat gland secretory duct through CFTR (red part).
 - In cystic fibrosis → less chloride reabsorption in the secretory duct (red part) → salty sweat.
- Sweat chloride testing can aid with diagnosis.

